# Ningna Wang

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I work in **geometry processing**, building **shape proxies**: compact representations of a 3D shape, from a skeleton that keeps only its topology to a distance field over space, each at the scale a task needs and with a guarantee on what it preserves, so that geometry can be trusted in **manufacturing**, **medicine**, **robotics**, and **urban planning**.

## Research Positions

**Postdoctoral Research Scientist, Columbia University** — New York, NY, USA
Advisor: Dr. Silvia Sellán — *2025 - present*

- Initiated the contact with **Meshy**, a major AI 3D generation company, that led to an industry-funded project I now lead, which funds my second postdoctoral year.
- Developing a series of robust methods to **extract surface mesh** from implicit fields.

**PhD Student, The University of Texas at Dallas** — Dallas, Texas, USA
Advisor: Dr. Xiaohu Guo — *2019 - 2025*

- Developed a complete framework for computing **high-quality** medial meshs of 3D models with **structure-awareness**.
- Developed a complete framework for computing the medial axis of 3D CAD meshes with **sharp-features preservation**.
- Developed a novel **topology-preserving** 3D medial axis computation framework based on volumetric restricted power diagram (RPD).
- Studied **learning-based methods** for mesh reconstruction via 3D skeleton prediction from **single-view images** or **point clouds**.
- Developed a new method for estimating **globally consistent normal orientations** for a raw point cloud.
- Studied a smooth **mesh simplification** functional that simultaneously consolidates weak features in a high-quality mesh.

**Research Intern, Shenzhen University** — Shenzhen, China
Advisor: Dr. Hui Huang — *10/2023 - 12/2023*

- Developed an **aerial path planning** method for high-quality digital 3D scene updation.

## Degrees

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|---|---|---|
| Ph.D. in Computer Science | **University of Texas at Dallas** \| Dallas, Texas, USA | *2025* |
| M.S. in Computer Science | **Carnegie Mellon University** \| Pittsburgh, PA, USA | *2016* |
| B.S. in Computationl Mathematics | **Jilin University** \| Changchun, Jilin, China | *2014* |

## Awards

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|---|---|---|---|
| *2025* | **EECS Rising Stars** | Massachusetts Institute of Technology (MIT) | USA |
| *2025* | **Rising Stars in Computer Graphics** | ACM WiGRAPH | USA |
| *2023* | **SIGGRAPH Technical Best Paper Award** | ACM SIGGRAPH | USA |
| *2025* | **CVM 2025 Best Paper Award** | Computational Visual Media Conference | HK, China |
| *2024* | **Betty and Gifford Johnson Travel Awards** | The University of Texas at Dallas | USA |
| *2024* | **First Place CAST STAR Award (1/16 teams)** | CAST-STEM Bridge Summer Camp | USA |
| *2013* | **Honorable Mention** | Mathematical Contest in Modeling (MCM) | USA |
| *2010* | **First/Second Prize Scholarship & Outstanding Student** | Jilin University (2010-2014) | China |

## Publications

[Journal & Conference Articles] * denotes equal contributions

[1] DC-SQEM: Medial Mesh Extraction by Dual Contouring with Spherical Quadric Error Metric **[Journal]**
Bingchuan Li, Ruilong Liu, Tianhao Hu, Congyi Zhang, **Ningna Wang**, Junfeng Yao, Xiaohu Guo
***to appear in ACM Transactions on Graphics (SIGGRAPH North America 2027)***, 2026.

[2] Dual Contouring of Signed Distance Data
Xiana Carrera, **Ningna Wang**, Christopher Batty, Oded Stein, Silvia Sellán
***ACM SIGGRAPH North America***. Los Angeles, USA, 2026.

[3] MVFusion-GS: Motion-Variance Guided Temporal Attention for High-Quality Dynamic Gaussian Splatting
Jianwei Hu, Tingxuan Huang, Hengyu Zhou, **Ningna Wang**, Xiaohu Guo, Jinshan Lai, Bin Wang
***European Conference on Computer Vision (ECCV)***, 2026.

[4] MATStruct: High-quality Medial Mesh Computation via Structure-aware Variational Optimization
**Ningna Wang**, Rui Xu, Yibo Yin, Zichun Zhong, Taku Komura, Wenping Wang, Xiaohu Guo
***ACM SIGGRAPH Asia***. Hong Kong, China, 2025: pp. 1–12.

[5] TexHOI: Reconstructing Textures of 3D Unknown Objects in Monocular Hand-Object Interaction Scenes **[Journal]**
Alakh Aggarwal, **Ningna Wang**, Xiaohu Guo
***IEEE Transactions on Visualization & Computer Graphics (TVCG)***, 2025, 31(10): pp. 8347–8358.

[6] Aerial Path Online Planning for Urban Scene Updation
Mingfeng Tang*, **Ningna Wang***, Ziyuan Xie, Jianwei Hu, Ke Xie, Xiaohu Guo, Hui Huang
***ACM SIGGRAPH North America***. Vancouver, Canada, 2025: pp. 1–11.

[7] CADDreamer: CAD Object Generation from Single-view Images
Yuan Li, Cheng Lin, Yuan Liu, Xiaoxiao Long, Chenxu Zhang, **Ningna Wang**, Xin Li, Wenping Wang, Xiaohu Guo
***Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)***. Nashville, USA, 2025.

[8] RBF-MAT: Computing medial axis transform from point clouds by optimizing radial basis functions **[Journal]**
Mengyuan Ge, Junfeng Yao, Baorong Yang, **Ningna Wang**, Zhonggui Chen, Xiaohu Guo
***Computer Aided Geometric Design (CAGD)***, 2025, 118(C): p. 102421.

[9] MATTopo: Topology-preserving Medial Axis Transform with Restricted Power Diagram **[Journal]**
**Ningna Wang**, Hui Huang, Shibo Song, Bin Wang, Wenping Wang, Xiaohu Guo
***ACM Transactions on Graphics (SIGGRAPH Asia)***, 2024, 43(6): pp. 1–16.

[10] NASM: Neural Anisotropic Surface Meshing
Hongbo Li, Haikuan Zhu, Sikai Zhong, **Ningna Wang**, Cheng Lin, Xiaohu Guo, Shiqing Xin, Wenping Wang, Jing Hua, Zichun Zhong
***ACM SIGGRAPH Asia***. Tokyo, Japan, 2024: pp. 1–12.

[11] CWF: Consolidating Weak Features in High-quality Mesh Simplification **[Journal]**
Rui Xu*, Longdu Liu*, **Ningna Wang**, Shuangmin Chen, Shiqing Xin, Xiaohu Guo, Zichun Zhong, Taku Komura, Wenping Wang, Changhe Tu
***ACM Transactions on Graphics (SIGGRAPH North America)***, 2024, 43(4): pp. 1–14.

[12] S3DS: Self-supervised Learning of 3D Skeletons from Single View Images
Jianwei Hu, **Ningna Wang**, Baorong Yang, Gang Chen, Xiaohu Guo, Bin Wang
***Proceedings of the 31st ACM International Conference on Multimedia (ACM MM)***. Ottawa, Canada, 2023: pp. 6948–6958.

[13] Point2MM: Learning medial mesh from point clouds **[Journal]**
Mengyuan Ge, Junfeng Yao, Baorong Yang, **Ningna Wang**, Zhonggui Chen, Xiaohu Guo
***Computers & Graphics (C&G)***, 2023, 115(C): pp. 511–521.

[14] Globally Consistent Normal Orientation for Point Clouds by Regularizing the Winding-Number Field **[Best Paper Award]**
Rui Xu, Zhiyang Dou, **Ningna Wang**, Shiqing Xin, Shuangmin Chen, Mingyan Jiang, Xiaohu Guo, Wenping Wang, Changhe Tu
***ACM Transactions on Graphics (SIGGRAPH North America)***, 2023, 42(4): pp. 1–15.

[15] Computing Medial Axis Transform with Feature Preservation via Restricted Power Diagram **[Journal]**
**Ningna Wang**, Bin Wang, Wenping Wang, Xiaohu Guo
***ACM Transactions on Graphics (SIGGRAPH Asia)***, 2022, 41(6): pp. 1–18.

[16] IMMAT: Mesh Reconstruction from Single View Images by Medial Axis Transform Prediction **[Journal]**
Jianwei Hu, Gang Chen, Baorong Yang, **Ningna Wang**, Xiaohu Guo, Bin Wang
***Computer-Aided Design (CAD)***, 2022, 150: p. 103304.

# Invited Talks

**Contouring and Operations on Implicit Geometry**
nTop | New York, NY, USA *06/2026*

**Inside Shapes: Understanding Shapes through Medial Geometry**
University of Toronto | Toronto, Canada *04/2026*
New England Symposium on Graphic | MIT, Cambridge, MA, USA *04/2026*
GAMES Webinar | Online *04/2026*

**Computing 3D Medial Axis Transform via Restricted Power Diagram (RPD)**
Visual Computing Research Center, Shenzhen University | Shenzhen, China *12/2025*

**MATStruct: High-Quality Medial Mesh Computation via Structure-aware Variational Optimization**
ACM SIGGRAPH Asia 2025 | HongKong, China *12/2025*
Center for Digital Media Computing, Xiamen University | Online *12/2025*

**Aerial Path Online Planning for Urban Scene Updation**
ACM SIGGRAPH 2025 | Vancouver, Canada *08/2025*

**MATTopo: Topology-preserving Medial Axis Transform with Restricted Power Diagram**
ACM SIGGRAPH ASIA 2024 | Tokyo, Japan *12/2024*
Visual Computing Research Center, Shenzhen University | Shenzhen, China *11/2023*
Center for Digital Media Computing, Xiamen University | Online *10/2024*

**Computing Medial Axis Transform with Feature Preservation via Restricted Power Diagram**
ACM SIGGRAPH ASIA 2022 | Daegu, South Korea *12/2022*
Center for Digital Media Computing, Xiamen University | Online *01/2024, 11/2022*

## Other Work Experience

**Booking.com B.V.** Amsterdam, Netherlands
Senior Software Engineer *2018 - 2019*
Software Engineer *2017 - 2018*

- Designed, built, and operated the **hotel availability search system**, serving thousands of search requests per second.
- Improved system stability and scalability by distributing hotel availabilities with **jump consistent hashing**.
- Collaborated with product and partner engineers to develop and deliver the search system.

**The Priceline Group Inc.** Amsterdam, Netherlands | Seattle, WA, USA
Graduate Software Engineer *2016 - 2017*

- Developed a **hotel inventory management system** for property listing, yield management, and revenue analytics.
- Implemented and tested new features for the **Genius loyalty program**, including discounts and travel rewards.

## Teaching

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|---|---|---|
| Mentor | MIT Summer Geometry Initiative | *2026 Summer* |
| Teaching Assistant | UTD CS4361/CS6366 Computer Graphics | *2025 Spring, 2024 Fall, 2021 Spring* |
| Teaching Assistant | UTD Clark Summer Research Program | *2024 Summer* |
| Teaching Assistant | CAST-STEM Bridge Summer Camp | *2024 Summer* |

- Supervisor of a 9-high-school-students team on a 6-weeks 3D talking face project.
- Instructor of 2-weeks full-day deep learning tutorials including lectures, coding demos and exercises.

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|---|---|---|
| Teaching Assistant | UTD CS6323 Computer Animation and Gaming | *2022 Fall* |
| Teaching Assistant | UTD CS4347 Database Systems | *2022 Fall, 2021 Spring* |
| Teaching Assistant | UTD CS6334 Virtual Reality | *2020 Spring* |
| Teaching Assistant | UTD CS4332 Introduction to Programming Video Games | *2019 Fall* |

## Academic Service

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| Conference | Program Committee | **ACM SIGGRAPH North America** | *2027* |
| Conference | Posters Jury | **ACM SIGGRAPH North America** | *2026* |
| Conference | Program Committee | **Pacific Graphics** | *2026* |
| Conference | Program Committee | **Symposium on Solid and Physical Modeling (SPM)** | *2026* |
| Conference | Program Committee | **Computer Graphics International (CGI)** | *2025* |
| Conference | Program Committee | **International Conference on Computational Visual Media (CVM)** | *2024* |
| Conference | Reviewer | **ACM SIGGRAPH North America** | *2026, 2025, 2024* |
| Conference | Reviewer | **ACM SIGGRAPH Asia** | *2026, 2025, 2024* |
| Conference | Reviewer | **Pacific Graphics** | *2026, 2023* |
| Conference | Reviewer | **Eurographics** | *2026* |
| Conference | Reviewer | **International Conference on 3D Vision (3DV)** | *2026* |
| Conference | Reviewer | **International Conference on Geometric Modeling and Processing** | *2024* |

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|---|---|---|---|
| Journal | Reviewer | **IEEE Transactions on Visualization and Computer Graphics (TVCG)** | *2026, 2025, 2022* |
| Journal | Reviewer | **Computer Aided Geometric Design (CAGD)** | *2024* |
| Journal | Reviewer | **Graphical Models (GMOD)** | *2025, 2024* |
| Journal | Reviewer | **IEEE Computer Graphics and Applications** | *2026* |

## Research Mentoring

I co-mentored these students with their own faculty advisors on research projects (theirs or mine); all papers are co-authored with me.

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| **Xiana Carrera** | PhD student, Columbia University · ***Papers: SIGGRAPH 2026; 2 under review*** | *2025–present* |
| **Beatrice Stotz** | Undergraduate student, Columbia University · ***Papers: 1 short paper under review*** | *2025–present* |
| **Eva Kato** | Visiting student, Columbia University; now Braid Technologies · ***Papers: SCF 2026 (poster)*** | *2025* |
| **Zilong Wang** | PhD student, The University of Texas at Dallas · ***Papers: 1 under review*** | *2025–present* |
| **Bingchuan Li** | PhD student, Xiamen University · ***Papers: TOG 2026*** | *2025–present* |
| **Ruilong Liu** | MS 2025, now PhD student, Xiamen University · ***Papers: TOG 2026*** | *2025–present* |
| **Tianhao Hu** | MS student, Xiamen University · ***Papers: TOG 2026*** | *2025–2026* |
| **Mingfeng Tang** | MS student, Shenzhen University · ***Papers: SIGGRAPH 2025*** | *2024–2026* |
| **Ziyuan Xie** | MS student, Shenzhen University · ***Papers: SIGGRAPH 2025*** | *2024–2025* |
| **Yibo Yin** | BS 2025, Wuhan University; now MS student, EPFL · ***Papers: SIGGRAPH Asia 2025*** | *2024–2025, 2026–present* |
| **Mengyuan Ge** | PhD student, Xiamen University · ***Papers: C&G 2023, CAGD 2025*** | *2022–2024* |
| **Shibo Song** | MS 2023, Xiamen University; then miHoYo · ***Papers: ACM TOG 2024*** | *2021–2024* |

## Software

Additionally, open-source code is available for publications above at https://ningnawang.github.io/.

**LibMAT** github.com/ningnawang/libmat

A C++ library for processing the 3D medial axis transform (MAT), with an emphasis on medial meshes.

**Blender-mat-addon** github.com/songshibo/blender-mat-addon

A Blender plugin for importing and exporting medial meshes (MA format) and interpolating spheres, cones, and slabs.

**Q-MAT open-source** github.com/ningnawang/QMAT

Open-source C++ code for the SIGGRAPH 2015 paper *Q-MAT: Computing Medial Axis Transform by Quadratic Error Minimization.*

**References available upon request.**